Greening Islington: Pocket Park Designs for Arsenal

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1 Introduction

1.1 Introduction

Climate risk, resilience and urban greening

Climate change continues to pose an existential threat to society and infrastructure, with its impacts felt most acutely by economically deprived communities. The impacts of the ecological crisis are compounded in urban environments due to the absence of green space and prevalence of hard landscaping. (Tan, Lau & Ng, 2016). In response, there are policy and planning efforts at the international, municipal, and local levels to achieve sustainable urbanisation and climate resilience in cities through green infrastructure and urban greening initiatives, such as pocket parks, sustainable urban drainage systems (SuDs), and green roofs.

There is a growing movement to catalyse trees and other forms of nature in urban spaces through development of “Tree Sensitive Urban Design”, as the quality of the urban tree canopy is a strong indicator of liveability in cities (Ely, 2008; Moskell & Allred, 2013). These interventions are essential to build climate-resilient, people-centred neighbourhoods, which align with the United Nations Sustainable Development Goals (SDGs) and fulfill obligations of the London Plan (Greater London Authority, 2021). This report examines the utility and tradeoffs of these interventions and policies with a focus on the potential of pocket parks within the broader urban greening agenda.

Figure 1: Sustainbale Development Goals (European Commission, n.d)

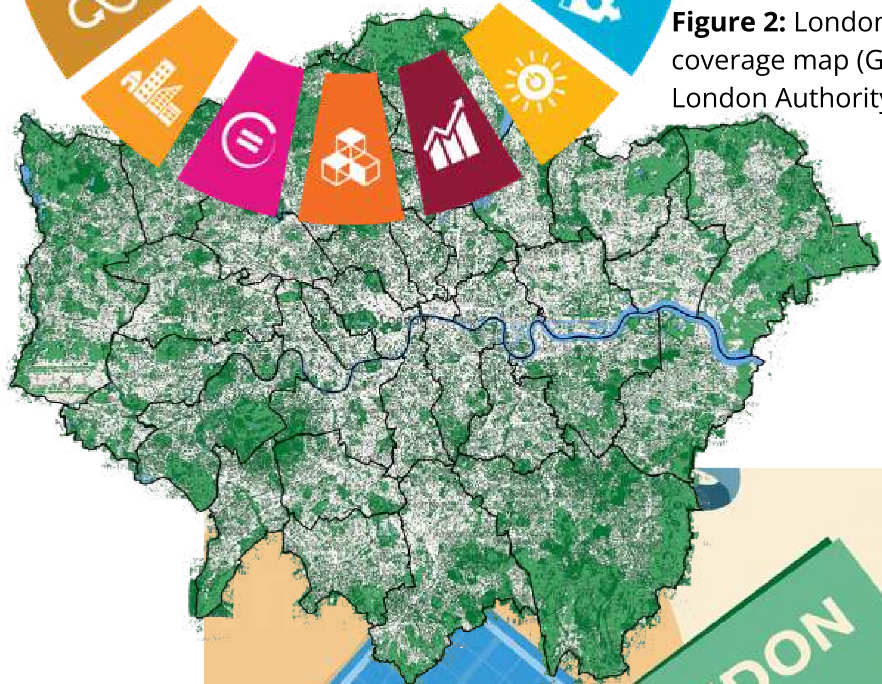


Figure 2: London green coverage map (Greater London Authority, 2025)

Figure 3: The London Plan (Greater London Authority, 2025).

1.2 Introduction

The context of the London borough of Islington

The London Borough of Islington is actively pursuing a "greener, healthier borough" through a range of climate policies and small-scale greening initiatives. Like other municipalities, the borough faces many climate and health challenges due to impacts of climate change, such as air pollution and extreme weather, resulting in higher flood and heat risk. Islington is one of the most densely populated local authority areas in England and Wales, with 16,321 people per square kilometre–almost triple the London average and more than 37 times the national average (State of Equalities Islington, 2021). Additionally, the borough is characterised by extreme disparities in deprivation and income, with some residents stating there are "Two Islingtons" (Islington Fairness Commission, 2011). This density is reflected in a stark lack of green space and tight-knit urban morphology as Islington has the lowest amount of green space per person of any local authority in England, with just 13% of the borough classified as public green space (Islington Local Plan, 2023). Furthermore, Highbury Fields, the borough's largest open green space, is just 11.75 hectares, compared to the 320 hectares of Hampstead Heath in the neighbouring borough of Camden (Islington Council, Heath Hands).

In response, the Islington Local Plan (2023) sets out a series of ambitious targets, including planting 600 more trees than are lost annually and achieving 1.5 hectares of new public green space by 2030. The borough also launched the Islington Greener Together initiative that invited residents to apply for funding to improve their local green spaces and, in 2024, launched its own Climate Panel composed of a diverse group of residents to inform local climate resilience (Islington Council, 2025). These programmes are demonstrative of the participatory efforts Islington Council takes to incorporate the public in climate and environmental policy, though stewardship for pocket parks remains a challenge. Beyond their environmental function, these interventions carry significant co-benefits for residents, with growing evidence of their capacity to improve physical health, mental well-being, and overall quality of life in dense urban settings (Turcu, 2025)

Figure 4: Example of urban greening implemented at Robert Blair Primary School (Islington Council, 2023).

ISLINGTON
For a more equal future

Figure 5: Islington logo (Islington Council, n.d.)

Figure 6: Parkscapes

1.3 Islington Site Analysis

Dense urban morphology
Due to a compact urban fabric, pocket parks and other small-scale interventions are the most viable greening options for the borough to achieve its climate and health goals.

Figure 7: Figure-ground map (Ordnance Survey, 2024)

Existing green coverage and tree canopy
This map illustrates Islington's green coverage and tree canopy. Comparing this with the figure ground map (Figure 7), one can see how the built environment poses a challenge to increase greening.

Figure 9: Green coverage and tree canopy map (Greater London Authority, 2023)

Deprivation: "Two Islingtons"
This map captures the varying inequalities in the borough, reflecting the close proximity of individuals experiencing very different levels of income and access to opportunities and resources.

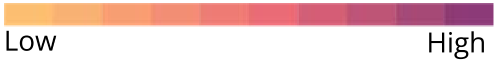


Figure 8: Index of multiple deprivation map (Ministry of Housing, Communities & Local Government, 2019)

Accessible open green space
In contrast, the actual accessible green space is much more limited, revealing that a large proportion of green coverage is from private gardens with implications for green space equity.

Figure 10: Accessible green space map (Natural England, 2025)

1.4 Arsenal's Demographic Context

In 2006, the Arsenal Emirates Football stadium was constructed as a regeneration scheme within Arsenal ward and larger Islington, providing new homes and funding for public space as part of the development (Webber, 2021). A cultural and economic cornerstone of Islington, the completion of the stadium brought an influx of investment to the area, but also resulted in rising property values, gentrification, and socio-economic inequality (Afgheldt and Kavetsos, 2010). This has led to over 46% of households within the ward remaining deprived as defined by the Office of National Statistics (See figure 13). Arsenal's location between some of wealthiest wards in Islington, namely Highbury, and some of the poorest, such as Finsbury Park and Highbury Road, makes it spatially representative of the socio-economic disparity that plays out across the borough as a whole. As seen in figure 12, deprivation within Arsenal is focused near major transit arteries such as A1 (Holloway Road) and the railway forming part of London's Overground.

The population in Arsenal is highly diverse, with nearly 40% of residents born outside of the UK (See figure 13). Additionally, 19.3% of the ward's population are under the age of 20, making it a particularly young neighbourhood. Since youth and the elderly are especially vulnerable to air pollution and extreme heat, urban greening interventions that target areas where they reside have the potential to mitigate lifetime health side-effects (Otto, et al. 2017). Additionally, economically and socially marginalised communities are most unable to cope with climate extremes while more likely to have less green space access compared to other socio-demographic groups (Otto, et al. 2017). Urban greening in Arsenal can maximise its impact by targeting areas with concentrated deprivation.

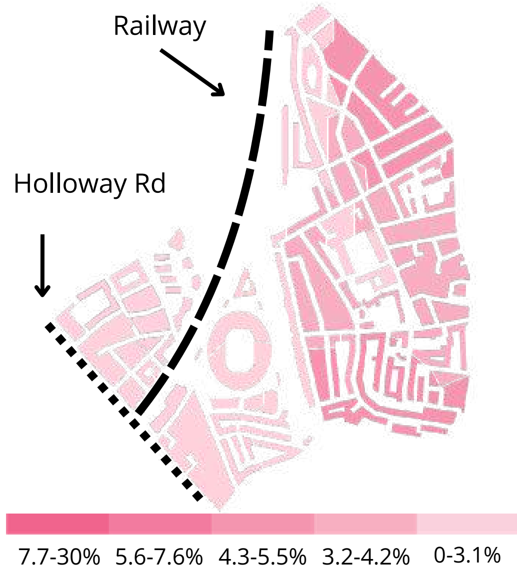


Figure 11. Age 75+ map
(Office for National Statistics, 2021)

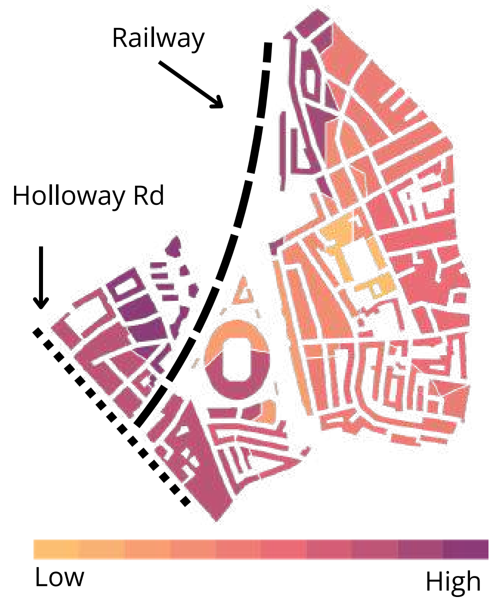


Figure 12. Deprivation index map
(MHCLG, 2019)

Figure 13: Demographic charts
(Office for National Statistics C, 2021)

1.5 Arsenal’s Environmental Context

Arsenal’s demographics illustrates a high potential for urban greening interventions to address overlapping environmental, health, and social challenges in the ward’s dense urban setting. Comparing the distribution of deprivation in Arsenal (Figure 12) to climate risks such as heat (figure 15) and flooding (figure 17), one begins to see a correlation between social, economic, and environmental vulnerability.

In particular, the large amount of vehicular traffic along A1 and brought by the Emirates stadium increases concentrations of NO2 and PM2.5 in areas with higher numbers of deprived residents (Figure 14 and 16). Surface temperature peaks in the North end of the ward towards Finsbury Park (Figure 15) while flood risk is highest in areas with larger amount of impermeable surface towards the West a (Figure 17).

As demonstrated in Figures 18 and 19, greenspace and tree canopy coverage in Arsenal is concentrated in private gardens and Gillespie Park, the largest publicly accessible green space in Arsenal (Islington, 2026 and Green Space Information for Greater London, 2025). However, the park’s location limits accessibility for residents on the opposite side of the railway with many living more than a 20 minute walk away. The high prevelance of private gardens also diminishes greenspace equity for for deprived residents.

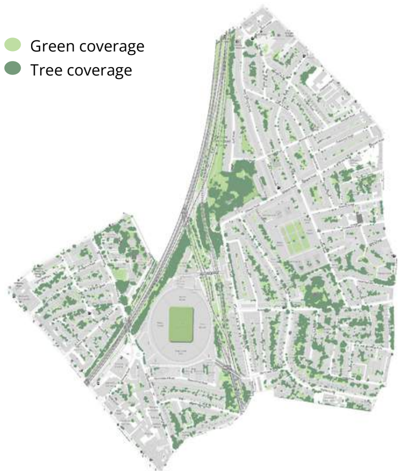


Figure 18. Arsenal green coverage (Greater London Authority, 2023)

Figure 19 . Arsenal accessible green space (Natural England, 2025)

Climate risk and green space maps

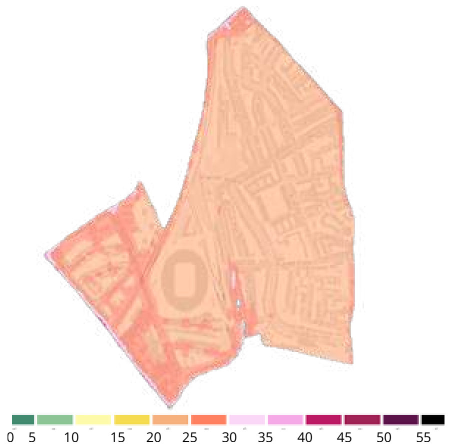


Figure 14: Mean annual NO2 concentrations (μg/m^3) 2022 (Greater London Authority, 2024)

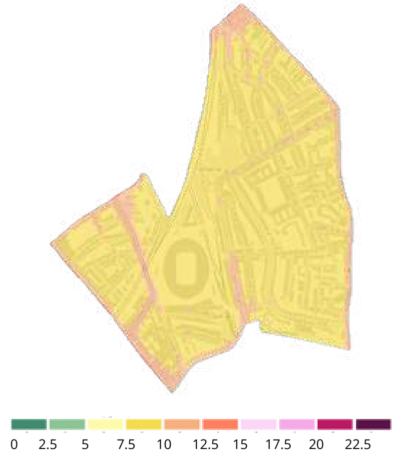


Figure 16: Mean annual PM2.5 concentrations (μg/m^3) 2022 (Greater London Authority, 2024)

Figure 15. Surface temperature map (Greater London Authority, 2020)

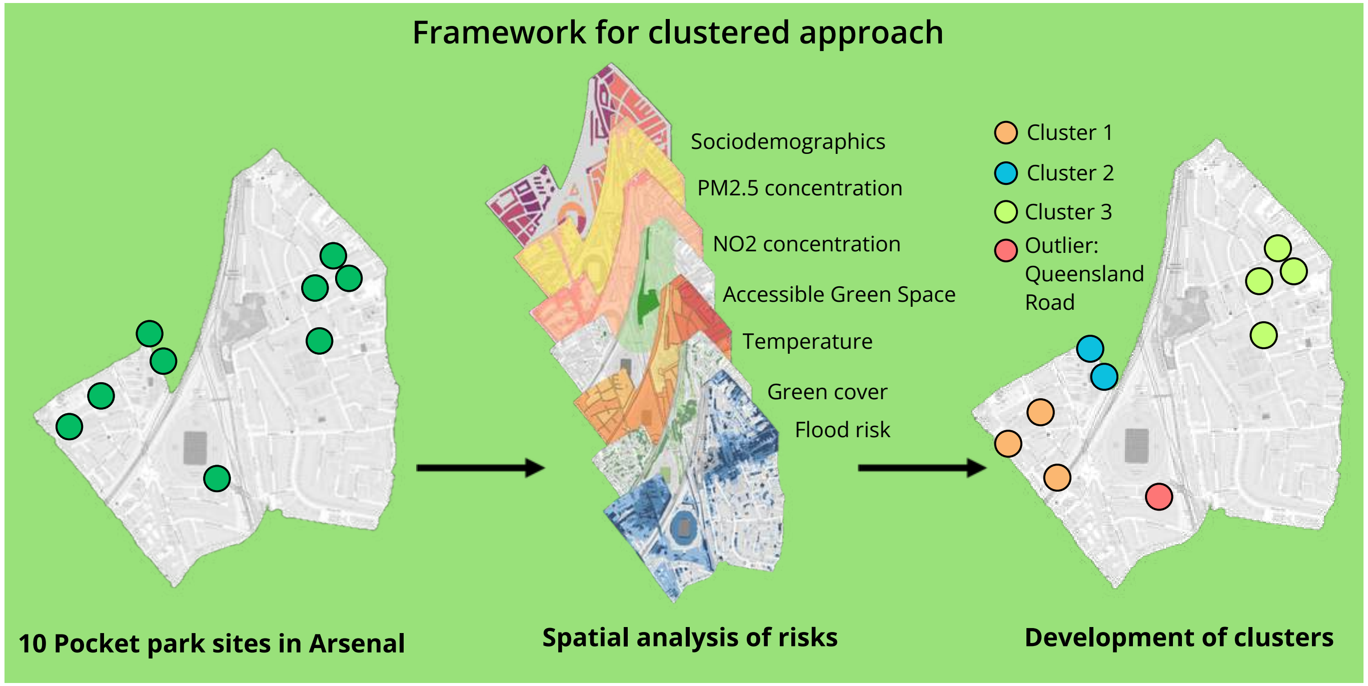
Figure 17. Arsenal flood risk map (Environment Agency, 2024)

2 Methodology

2.1 Methodology

Sites, clusters and typologies

This project outlines pocket park interventions on 10 sites across the ward of Arsenal, created following design guidance within the Green Finance of Islington Pocket Park Framework by WSP (2024). To identify the spatial patterns of climate and health risks within the ward, socio-demographic and climate data was gathered and mapped in QGIS (QGIS Development Team, 2026). To maximise climate and health benefits associated with pocket parks, a 'clustered approach' was adopted, whereby sites were grouped by common characteristics. Clusters were created based on geographic proximity, as well as shared climate and socio-demographic risks, allowing a more targeted approach to pocket park design that prioritises multifunctionality and co-benefits.



Calculating climate and health benefits

Estimates of the climate benefits of trees, namely carbon storage and sequestration, surface water runoff, and pollutant (NO2 and PM2.5) removal were calculated using the Islington i-Tree Eco Inventory Report on Public Trees (Watson and Hill, 2019). Accurately quantifying the health benefits resulting from reduced pollution is challenging, as it is difficult to isolate the impact of single trees in environments were numerous external factors are at play. However, the UK Department for Environment, Food and Rural Affairs' (DEFRA, 2025) Air Quality Appraisal tool was used to calculate the damage costs associated with the removal of pollutants from trees. The Urban Greening Factor (UGF) is a tool introduced by the London Plan Policy G5 that evaluates the quantity and quality of urban greening, and is now a fundamental element of site design (Greater London Authority, 2021). This tool was used to calculate the UGF before and after pocket park interventions.

Tree selection

Trees in London's public spaces play an important role for both people and wildlife in the city, but a changing climate is threatening the survival, longevity and resilience of urban trees (Hirons and Martin, 2025). Climate suitability differs by tree species and location, and London is one of the few cities with future climate suitability data on all (1.1 million) of its public realm trees (Greater London Authority, 2025). However, climate suitability is just one of the considerations in the process of selecting trees for pocket parks. Ecosystem services, such as carbon sequestration, also differ drastically by species (Watson and Hill, 2019). Larger trees yield higher ecosystem services, and space limitations are one of the most important factors in designing pocket parks given their small size. Additionally, characteristics such as thorns and allergenic pollen, can make a species unsuitable at the local-level. These tradeoffs were balanced by considering trees with the highest ecosystem services, future climate suitability and local suitability within spatial constraints.

3 Cluster 1:
Lowman Road, Dunford Road, and Annette Road

3.1 Cluster 1 Site Analysis

Site characteristics

Cluster 1, which includes the sites at Lowman Road, Dunford Road, and Annette Road, is located to the west of the Emirates Stadium. In contrast to the stadium's large-scale, managed green spaces, this cluster is embedded within a high-density residential neighborhood. The neighborhood features a younger demographic compared to the wider Arsenal area and experiences higher levels of deprivation.

Climate and health risks

Cluster 1's vulnerability to climate extremes is high, experiencing significant flood and moderate heat risks. Socially, these sites are entirely embedded within a corridor of intensified relative deprivation, reflecting a highly vulnerable demographic that faces systemic barriers to resources. The compounding of environmental hazards with socio-economic disadvantage demonstrates a critical alignment of climate and community needs. Due to these intersecting vulnerabilities, targeted interventions focused on enhanced urban greening and flood mitigation.

Figure 20. Lowman Road current planter boxes

Climate and health risk maps

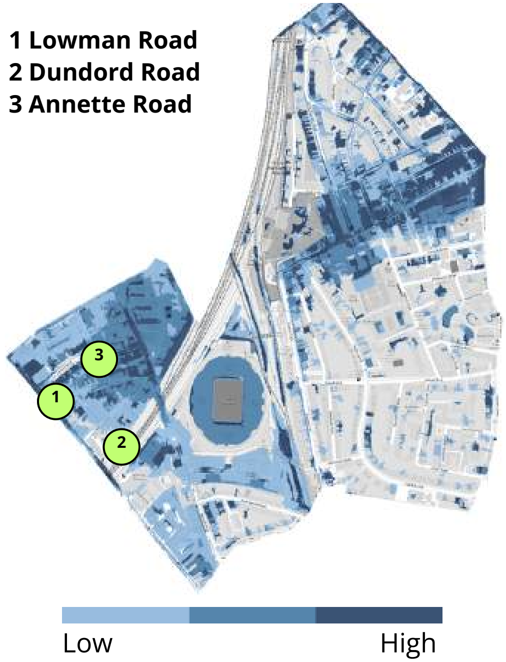


Figure 21. Arsenal flood risk map
(Environment Agency, 2024)

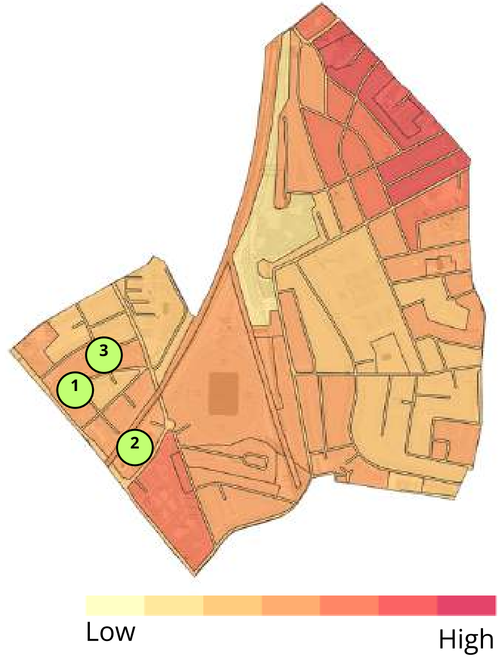


Figure 23. Surface temperature map
(Greater London Authority, 2020)

Figure 22 . Deprivation index map
(MHCLG, 2019)

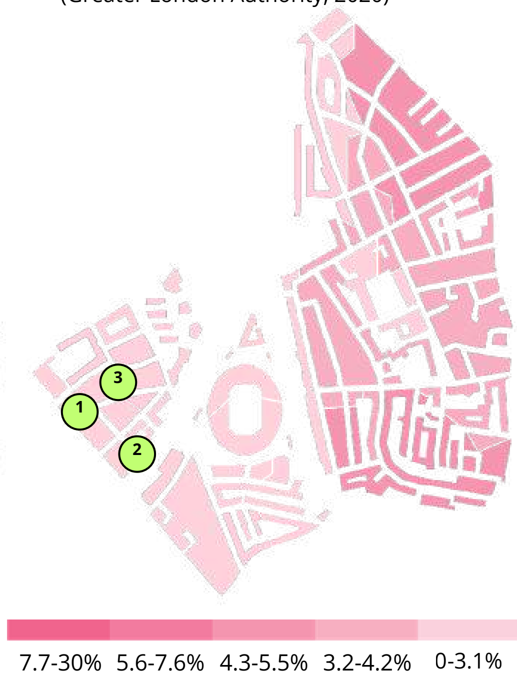


Figure 24. Age 75+ map
(Office for National Statistics, 2021)

3.2 Cluster 1 Tree Canopy and Green Coverage

Existing tree canopy and green coverage

Cluster 1 experiences significantly lower existing tree canopy compared to the wider Arsenal area. As illustrated in Figure 22, Cluster 1 is characterised by sparse and highly fragmented vegetation. The spatial isolation worsens green space inequity, especially for households lacking private gardens (See Figure 23). While the dense urban fabric restricts large-scale green infrastructure, targeted micro-interventions are highly viable.

Funding and stewardship

While comprehensive modifications are currently proposed under the Annette Road Area Liveable Neighbourhood (Islington Council, 2026) programme, an alternative strategy emphasising targeted micro-interventions across Annette, Lowman, and Dunford Roads is strongly advocated (See Figure 24). The approach offers a highly cost-effective solution, which is critical to ensuring equitable implementation given the neighborhood's deprivation rates. Local authority oversight remains the primary means of governance through Greener Together Initiative (Islington Council, 2024). Furthermore, the nearby Waitrose car park offers an additional site to contribute to this green network. Involving the supermarket as a corporate partner would diversify maintenance resources and strengthen the resilience of local green spaces.

Figure 25. Proposed Annette Road (Islington Council, n.d.)

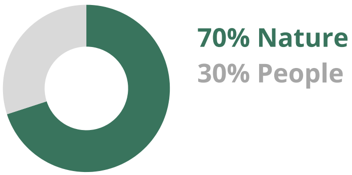
Figure 26. Location of Waitrose car park (Openstreet Map)

Figure 27. Arsenal green tree canopy and green coverage (Greater London Authority, 2023)

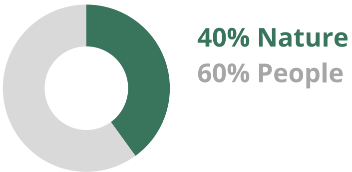
Figure 28. Arsenal accessible green space (Natural England, 2025)

3.3 Cluster 1 Pocket Park Designs

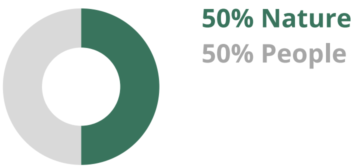
1 Lowman Road
Urban Bee Paradise



2 Dunford Road
Community Garden



3 Annette Road
Community Artscape / Active Transport



Flood Mitigation and Community Enhancement

Cluster 1 is characterised by a closed street layout, seen in the dead-end streets of Lowman Road and Dunford Road. The quiet, low-traffic layout provides the foundation for interventions that address flooding issues and enhanced community cohesion.

Figure 29. Cluster 1 pocket park in plan.

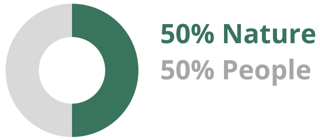
3.4 Annette Road Pocket Park Design

Figure 30. Urban art space at Annette Road

Figure 31. Existing Annette Road in plan

Urban Art Space

Given that there are already numerous trees on the site, the proposal prioritises enhancing existing space rather than adding tree canopy. Core design components include a newly added public seating alongside planter boxes which enhance the cycle lane. Furthermore, vibrant community artworks are integrated into existing facilities.



Design components

BENEFITS

Social and health: Stronger local identity vibrant space through community artwork

Climate and environment: Minimised construction and resource consumption through optimisation of existing green spaces.

3.4 Annette Road Pocket Park Design

Figure 32. Existing Annette Road (Google Maps, 2022)

Figure 33. Proposed Annette Road vision

3.5 Lowman Road Pocket Park Design

Existing Tree   New Tree

UGF: 0.54

Figure 34: Lowman Road pocket park design

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
1.324 tonnes	0.038 tonnes/year	1.245 m3/year	20 m	0.021 kg/yr	0.234 kg/yr

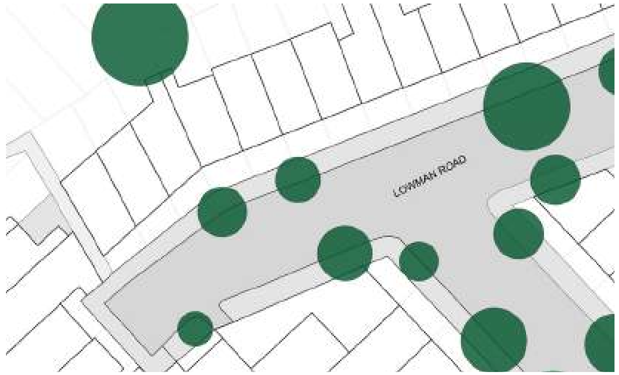


Figure 35: Existing Lowman Road in plan

Urban Bee Paradise

This design incorporates environmental policy efforts to rewild London through habitat creation and nature recovery, an ongoing priority for Mayor Sadiq Khan (Turnbull, Fry, and Lorimer, 2025).

Design components

With the addition of two wild cherry trees, one common lime trees and native flowers and other plantings, the end of Lowman Road is transformed into a haven for pollinators. Rocks to provide informal seating, lighting, and paths to connect home entrances to the street make this appealing to community members.

BENEFITS

Social and health: Improved mental health, connection to nature, nature-positive place making.

Climate and environment: Improve pollination, reduce flooding, habitat creation.

3.6 Dunford Road: A Community Garden for All

Existing Tree New Tree

Figure 37: Dunford Road pocket park designs

UGF: 0.68

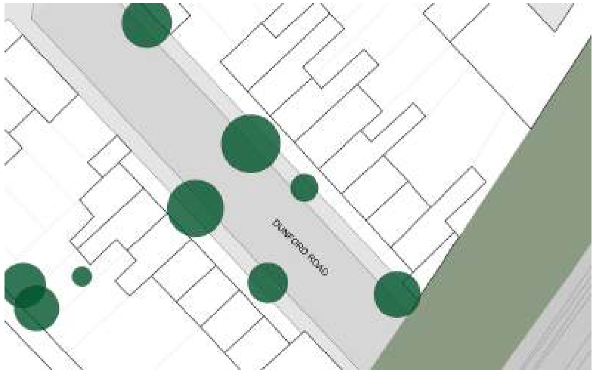


Figure 36: Exisiting Dunford Road in plan

Community Garden

Situated at a dead-end road in an area with a high concentration of elderly residents, this urban greening intervention is designed as a community garden to enhance the area's social infrastructure.

40% Nature
60% People

Design components

A picnic table, benches, and supply shed give users a place to gather while impermeable surface reserved for cars transforms into a community garden consisting of permeable materials. Two new deciduous trees provide shade in summer and lets in light throughout winter.

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
0.095 tonnes	0.4 tonnes/year	1.24 m3/year	23 m	0.018 kg/yr	0.3 kg/yr

BENEFITS

Social and health: Improved mental health, connection to nature, improved social connection, food production.

Climate and environment: Healthy planting, improved air quality, reduced flood risk

4 Cluster 2:
Citizen Road

4.1 Cluster 2 Site Analysis

Site characteristics

Citizen Road is a quiet, low-traffic dead-end street on the eastern side of the railway line in Holloway, N7, adjacent to the Harvist Estate, a social housing estate. The area is surrounded by residential buildings, and proximity to the Emirates Stadium brings periodic match-day pedestrian pressure. Overall, the site has strong location characteristics but any development would need to carefully address access, and noise from the railway.

Climate and health risks

Citizen Road sits within one of Islington's most deprived neighbourhoods according to Figure 40, a factor that adds to existing climate and health vulnerabilities for residents. Given the density of impermeable surfaces across the estate and limited capacity of urban drainage infrastructure, flood risk is high (see figure 39) leading to surface water flooding. This poses a risk to ground floor dwellings and shared spaces. Heat risk is low to medium and there is a small proportion of residents aged 75 and over. Overall, the intersection of climate vulnerability and social deprivation at this location points to a community with limited adaptive capacity.

Figure 38. Citizen Road

Climate and health risk maps

Figure 39. Arsenal flood risk map
(Environment Agency, 2024)

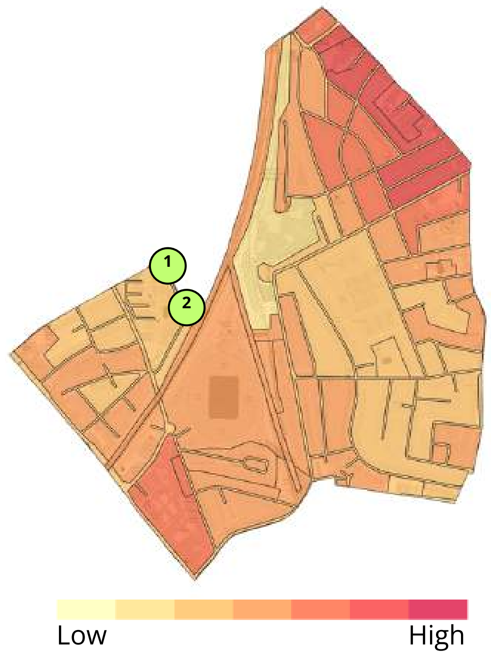


Figure 41. Surface temperature map
(Greater London Authority, 2020)

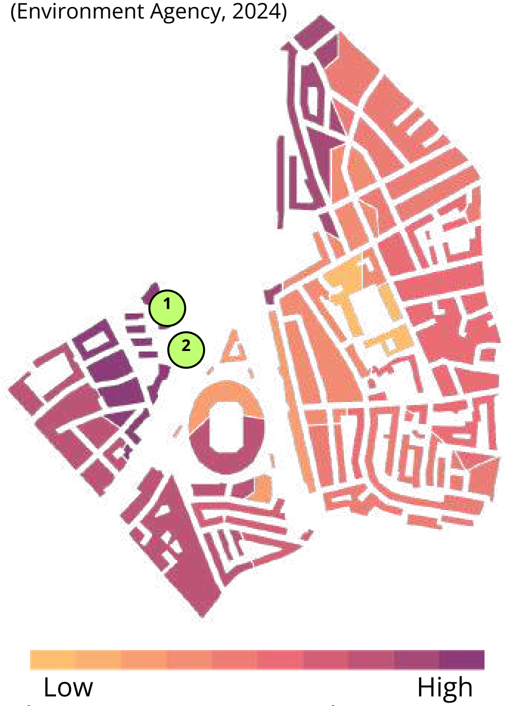


Figure 40. Deprivation index map
(MHCLG, 2019)

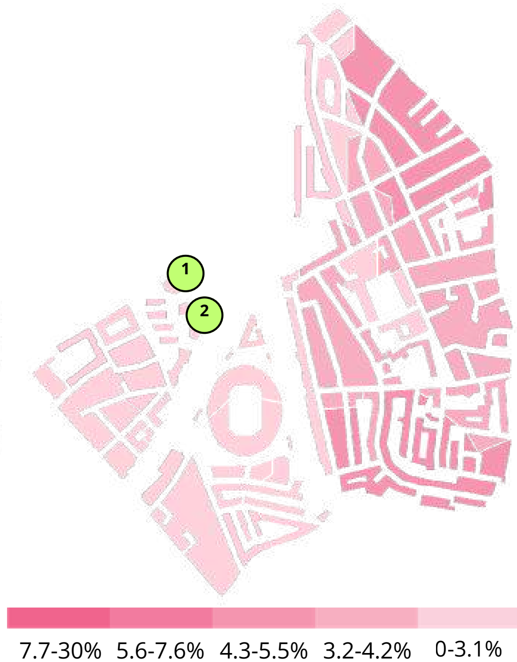


Figure 42. Age 75+ map
(Office for National Statistics, 2021)

4.2 Cluster 2 Tree Canopy and Green Coverage

Existing tree canopy and green coverage

Citizen Road has some existing green and tree canopy cover within the Harvist Estate setting, with approximately 207, mostly mature, trees of different species recorded across the estate (IslingtonGov, n.d.). Publicly accessible open space is limited locally, Gillespie Park is the closest green space which is approximately a 15-minute walk away, a distance that appears reasonable on paper but is undermined by the quality of the route. The walk requires passing under railway infrastructure, which creates an unpleasant pedestrian experience, and on Arsenal match days, the surrounding streets become heavily congested.

Stewardship and funding

Delivering and maintaining green infrastructure across the Citizen Road sites will require a clear stewardship and funding strategy. Given the context of an estate, Islington Council will be the lead partner with maintenance responsibilities being integrated into existing estate management contracts. Funding could be pursued through the UK Shared Prosperity Fund and the Mayor of London's Green and Resilient Spaces Fund (GLA, 2023). Community-led stewardship models, as stated by Natural England's green infrastructure framework (Natural England, 2023), can offer a complimentary approach by engaging local residents in the care of their local spaces whilst keeping individual burden low through council backing.

Scaling opportunities

Green infrastructure interventions work best when implemented as a connected network rather than in isolation to maximise health and ecological benefits (EEA, 2011). Beyond the two Citizen Road sites, there are other potential locations within and around the Harvist Estate that could benefit from pocket park interventions, helping to bridge the gap between the estate and Gillespie Park and creating a greener, more connected environment for residents. A key methodological reflection relates to pollution reduction calculations for NO2 and PM2.5. The absorption data available for the Paper Birch and Lawsons Cypress states values of less than 0.01 tonnes, meaning the figures used in the calculations are estimates rather than precise measurements.

Figure 43. Arsenal green tree canopy and green coverage (Greater London Authority, 2023)

Figure 44. Arsenal accessible green space (Natural England, 2025)

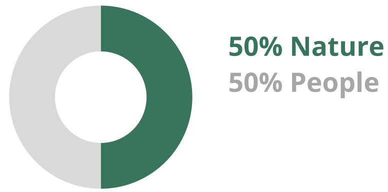
4.3 Cluster 2 Pocket Park Designs

Figure 45. Proposed Cluster 2 pocket park network

Citizens’ plot to enhance community

Proposed designs for Cluster 2 foster social connections and community resilience through accessible, activity-led open green spaces. A community garden typology provides residents with opportunities to grow food and engage with neighbours which is complemented by a spontaneous play typology, incorporating natural and informal play features that encourage unstructured outdoor activity. The overall approach prioritises community ownership and everyday use, reflecting the needs of a diverse residential population. Given that the second site on Citizen Road had spatial limitations, the proposal suggests an extension to the site connecting both sites through the adventure playground.

1 Citizen Road I
Community Garden



2 Citizen Road II
Adventure Playground

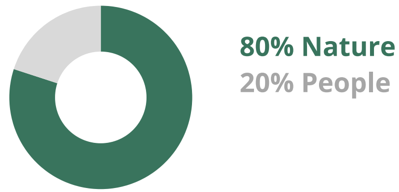


Figure 46. Existing Citizen Road

4.4 Citizen Road Pocket Park Design I

Community Garden

Social Garden WSP typology designed at the end of Citizen Road to provide a space for the local community to engage in social activities that also entail health and other hidden benefits.

50% Nature
50% People

Design components

BENEFITS

Social and health: Improved access to nature, supporting wellbeing

Climate and environment: Better flood management, reducing surface water runoff

Mental wellbeing: Physical and mental health benefits from community cohesion with added nutritional benefit, with locally grown produce supplementing access to fresh food.

Figure 47. Proposed Citizen Road I pocket park

Existing Tree New Tree

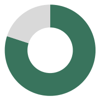
UGF: 0.38

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
0.14 tonnes	0 tonnes/year	0.09 m3/year	11m	<0.01 kg/yr	<0.01 kg/yr

4.5 Citizen Road Pocket Park Design II

Adventure Playground

Spontaneous Play WSP typology designed along Citizen Road as an integrated playground space for the younger residents of Harvist Estate. Low vehicle traffic on the road creates an ideal location for this typology.



80% Nature
20% People

Design components

BENEFITS

Social and health: Improved access to nature, supporting wellbeing.

Climate and environment: Better flood management, reducing surface water runoff and use of reclaimed materials to recycle and reduce waste.

Mental wellbeing: Unstructured play area for children, supporting motor development, physical activity and emotional resilience while fostering a connection to nature.

Figure 48. Proposed Citizen Road I pocket park

Existing Tree   New Tree

UGF: 0.43

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
2.75 tonnes	0.71 tonnes/year	2.17 m3/year	30-40m	0.03 kg/yr	0.348 kg/yr

5 Cluster 3: Plimsoll Road, Monsell Road, Chatterton Road and Elwood Street

5.1 Cluster 3 Site Analysis

Site characteristics

Cluster 3 includes sites on Plimsoll Road, Monsell Road, Chatterton Road, and Ellwood Street. It is in the Northeast of Arsenal, a residential area dominated by 2-3-storey terrace houses. Most households have access to a private rear garden, suggesting a lower need for recreational green space. The sites are located along local through roads (Figure 49), so closing off roads might not always be feasible and impacts of proposed typologies, on vehicular flows need to be considered. Major facilities include multiple primary schools, an arts school for children, and a childcare service centre, so many children likely use these roads on their way to school. Proposed green infrastructure interventions in this cluster serve as traffic-calming measures, increasing pedestrian safety for children (Dodd, 2024).

Climate and health risks

Heat and flood risks in this area are higher than in the rest of the ward (Fig.50, Figure 52).. The area experiences less deprivation than the rest of Islington, ranking as less deprived than 90% of the borough (GOV.UK, 2025) (Figure 51). Compared to the rest of the ward, it has a slightly higher proportion of residents aged 75 and over (Figure 55). As elderly and children are particularly vulnerable to negative health impacts from heat exposure, there is a need for heat mitigation (Taylor *et al.* 2024). Proposed GI designs in this cluster prioritise reducing heat, improving road safety, and reducing flooding from surface water runoff.

Figure 49. Google street view: Chatterton Road, Islington (Google Maps, 2026)

Climate and health risk maps

Figure 50. Arsenal flood risk map (Environment Agency, 2024)

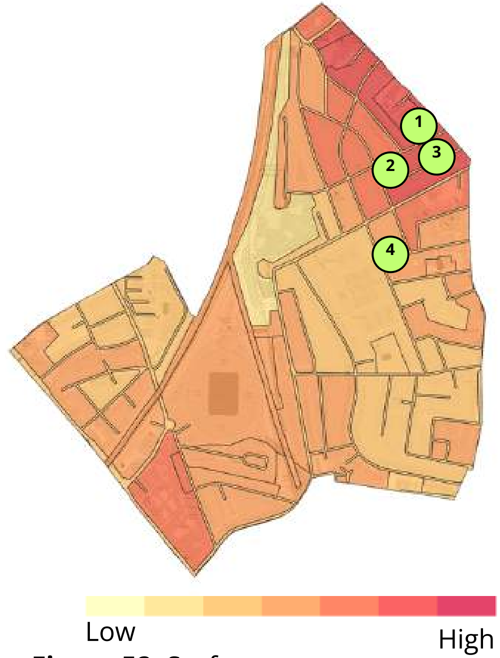


Figure 52. Surface temperature map (Greater London Authority, 2020)

Figure 51. Deprivation index map (MHCLG, 2019)

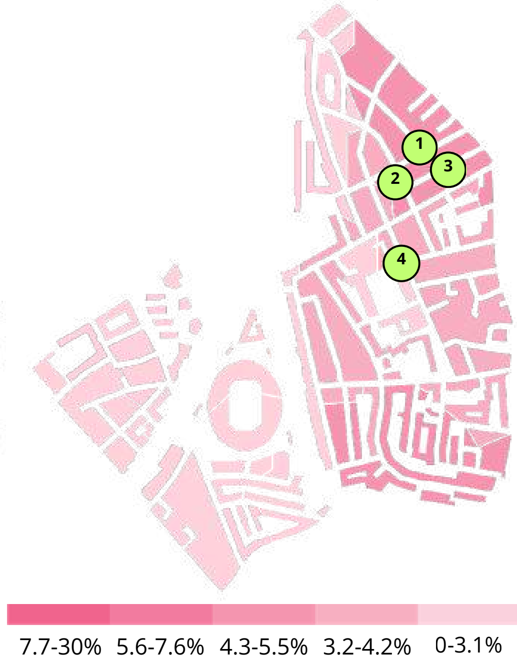


Figure 53. Age 75+ map (Office for National Statistics, 2021)

5.2 Cluster 3 Tree Canopy and Green Coverage

Existing tree canopy and green coverage

Although private gardens are common in the area (Figure 54), publicly accessible green space is not (Figure 55). Those without a garden may still be limited in their access to green space, and relying on Gillespie Park for public green space provision may limit equitable access for those living farther from the park or with mobility challenges. Cluster sites may not suit green space provision for recreational use but they could increase visual access and the perception of greenery by distributing green cover more evenly.

Stewardship and funding

Community involvement in this cluster is fundamental. As car ownership is higher here (ONS, 2021), residents might feel negatively affected by reductions in on-street parking spaces. Co-design and community stewardship can mitigate this and increase acceptance (Schreibmüller et al., 2025). The network of local schools may further support community stewardship, as parents' groups have shown keen interest and involvement in Islington's Greener Together (IGT) Champions stewardship programme in the past (Groundwork, n.d. a). Delivering the GI typologies as part of the IGT programme will allow for funding provision by Islington Council and the Future Parks Accelerator (Groundwork, n.d.).

Figure 54. Arsenal green tree canopy and green coverage (Greater London Authority, 2023)

Figure 55. Arsenal accessible green space (Natural England, 2025)

5.3 Cluster 3 Pocket Park Designs

Greening for pleasant commutes

The main priorities for GI typologies in this cluster are mitigating climate impacts and enhancing active travel. Nature-based typologies have been selected for their ability to regulate microclimates and support active travel modes. Newly planted trees will reduce heat by providing shade and decrease surface water runoff through water absorption (Taylor *et al.*, 2024). Depaved, planted areas on curb buildouts, which typically have fewer utilities underneath, can create a series of SUDs along roads while enhancing the streetscape (Dodd, 2024). Narrowing road sections will help reduce traffic, speeding and collisions (Dodd, 2024). Overall, typologies will support Islington's aim to encourage active travel modes (Islington, 2020) by reducing car dominance and creating a pleasant walking and cycling experience. This will not only improve air quality but also ensure better health outcomes for the local population and make transport more accessible for all (Islington, 2020).

Figure 56. Proposed network of cluster typologies

1 Plimsoll Road
Active Transport Landscape

80% Nature
20% People

2 Monsell Road
Active Transport Landscape

80% Nature
20% People

3 Chatterton Road
Active Transport Landscape

80% Nature
20% People

4 Ellwood Street
Play on the Way

50% Nature
50% People

5.4 Plimsoll Road Pocket Park Design

Figure 57. Active transport typology Plimsoll Road

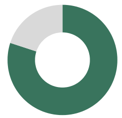
UGF: 0.44

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
0.7 tonnes	0.02 tonnes/year	0.89 m3/year	18 m	0.01 kg/yr	0.23 kg/yr

Figure 58: Existing Plimsoll Road in plan.

Active Transport Landscape

The active transport landscape typology encourages bike usage through the provision of new cycle storage. Curb buildouts with attractive planting enhances street charaacter, reduces surface water runoff through water absorption, and increases safety by closing the road section to cars. Newly planted trees provide shade and shelter from rain.



80% Nature
20% People

Design components

BENEFITS

Social and Health: Improved cycling experience, safer roads.

Climate and environment: Reduced surface water runoff, reduced air pollution, reduced heat and provision of shade.

Plimsoll Road Pocket Park Design

Figure 59. Plimsoll Road currently (Google Maps, 2026)

Figure 60. Plimsoll Road with proposed active transport typology

5.5 Monsell Road and Chatterton Road Pocket Park Designs

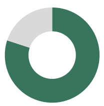
Active Transport Landscape

Existing Tree New Tree

Figure 61. Proposed active transport typology Monsell Road

UGF: 0.28

Design components



80% Nature
20% People

BENEFITS

Social and health: Improved active transport experience, safer roads.

Climate and environment: Reduced surface water runoff, reduced air pollution, reduced heat and provision of shade.

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	TREE CANOPY ADDED	PM 2.5 REDUCED	NO 2 REDUCED
0.74 tonnes	0.01 tonnes/yr	0.99 m3/yr	19 m	0.01 kg/yr	0.24 kg/yr

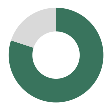
Active Transport Landscape

Existing Tree New Tree

Figure 62. Proposed active transport typology Chatterton Road

UGF: 0.36

Design components



80% Nature
20% People

BENEFITS

Social and health: Improved active transport experience, safer roads.

Climate and environment: Reduced surface water runoff, reduced air pollution, reduced heat and provision of shade

CARBON STORAGE	CARBON SEQUESTERD	RUNOFF AVOIDED	TREE CANOPY ADDED	PM 2.5 REDUCED	NO 2 REDUCED
1.14 tonnes	0.03 tonnes/yr	1.38 m3/yr	49 m	0.09 kg/yr	0.35 kg/yr

5.6 Elwood Street: Play on the Way

Figure 63. Proposed pocket park typology Ellwood Street

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
0.725 tonnes	0.39 tonnes/year	1.59 m3/year	29.8 m	0.0085 kg/yr	0.232 kg/yr

Figure 64. Existing Ellwood Street

Play on the Way

Located a block away from the the St John's Primary School, this urban greening intervention enhances Elwood Street through a "play on the way" design that prioritises pedestrian safety for students commuting to and from school.

50% Nature
50% People

Design components

Three new trees are added while bollards and an elevated permeable surface area closes the area off to vehicular traffic. Large stones located off the pedestrian path provide an opportunity for imaginative and spontaneous play.

BENEFITS
Social and health: Improved mental health, improved safety, improved social connection.

Climate and environment: Improved air quality, decreased flood risk.

6 Outlier: Queensland Road

6.1 Queensland Road Site Analysis

Site characteristics

Queensland Road is located directly south of Emirates Stadium, home to Arsenal football club, and comprises housing complexes that are a major component of the Arsenal Regeneration Area (CZWG, n.d.). There are also multiple vehicle entrances to the stadium (see Figure 65). This site is unique due to its lack of spatial connectivity to other sites and the amount of hard landscaping that cannot be changed due to the stadium's infrastructure. Thus, it exists outside of the green network established in this report, though the site offers a vital opportunity to improve tree planting around the stadium.

Climate and health risks

This site's climate and demographic needs don't share strong alignment with the other clusters as vulnerability to climate from extreme weather is moderate and it serves a more affluent demographic (Figures 66-69). Due to these factors and physical constraints, minimal interventions to increase urban greening and support climate-friendly transportation infrastructure are appropriate.

Figure 65. Queensland Road street view (Google Maps, 2025).

Climate and health risk maps

1 Queensland Road

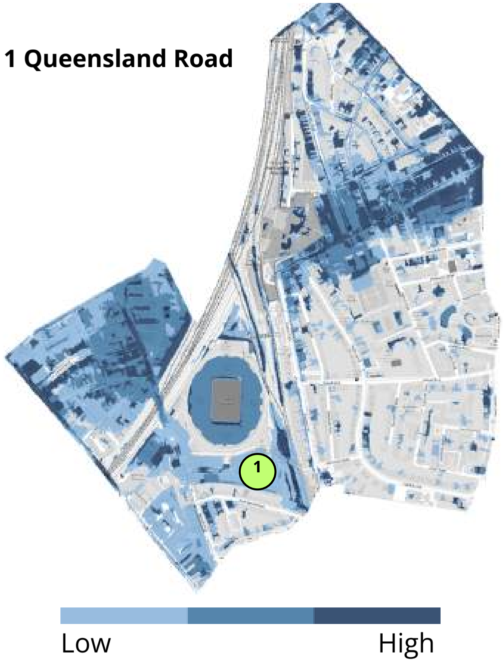


Figure 66. Arsenal flood risk map (Environment Agency, 2024)

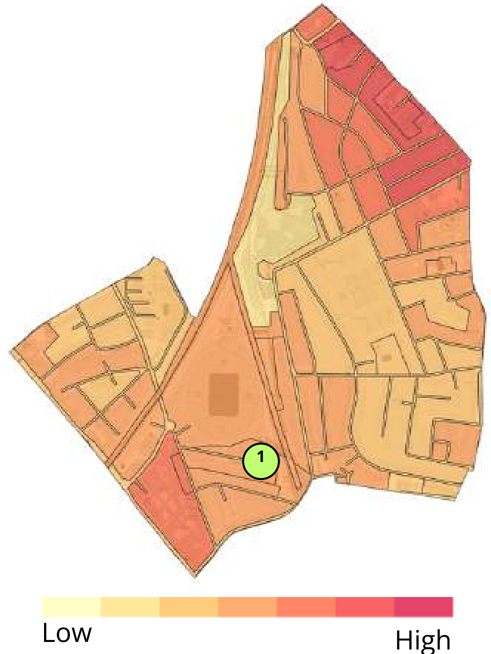


Figure 68. Surface temperature map (Greater London Authority, 2020)

Figure 67. Deprivation index map (MHCLG, 2019)

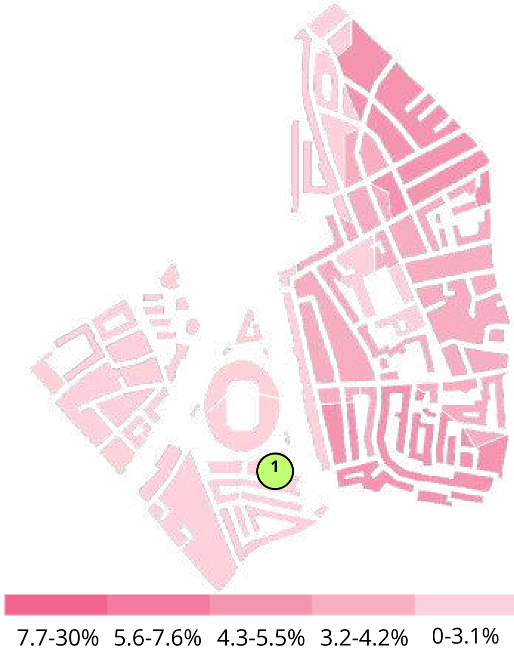


Figure 69. Age 75+ map (Office for National Statistics, 2021)

6.2 Queensland Road Tree Canopy and Green Coverage

Existing tree canopy and green coverage

There is low green coverage and tree canopy on Queensland Road in comparison to other streets in the ward. Specifically, there are no street trees besides a few planter boxes on the pavement (see Figure 70). The housing contains green roofs and two semi-private gardens for residents of the housing complex. Although it is adjacent to a dense tree canopy alongside the railway, this is inaccessible to the public (see Figure 71). Green infrastructure interventions are limited on the site due to the need for car access to the stadium and a large amount of disabled parking spaces on the street.

Stewardship and funding

This site offers a vital opportunity to extend the green network around Arsenal Emirates Stadium for increased climate benefits and improved stewardship. Since the Stadium and football club is cherished by the community, there are potential partnerships to educate and involve football fans in urban greening measures around Islington.

Figure 70. Queensland Road tree canopy and green coverage.

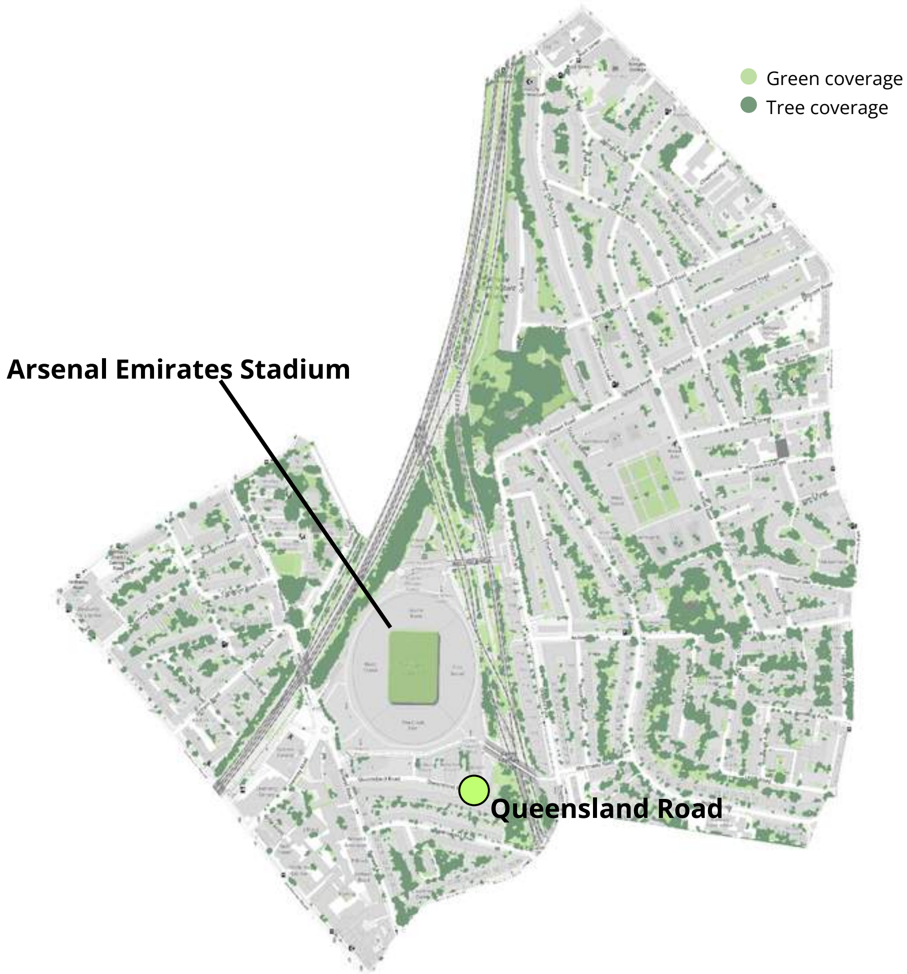
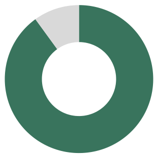


Figure 71. Arsenal tree canopy and green coverage around Queensland Road.

6.3 Queensland Road Pocket Park Design

Active Transport Landscape

An active transport typology will increase tree canopy cover and provide bike parking spaces for residents and community members. We propose planting two Norway maple trees and two common lime trees, expanding part of the pavement to allow for root growth and more tree-sensitive design (Ely, 2008).



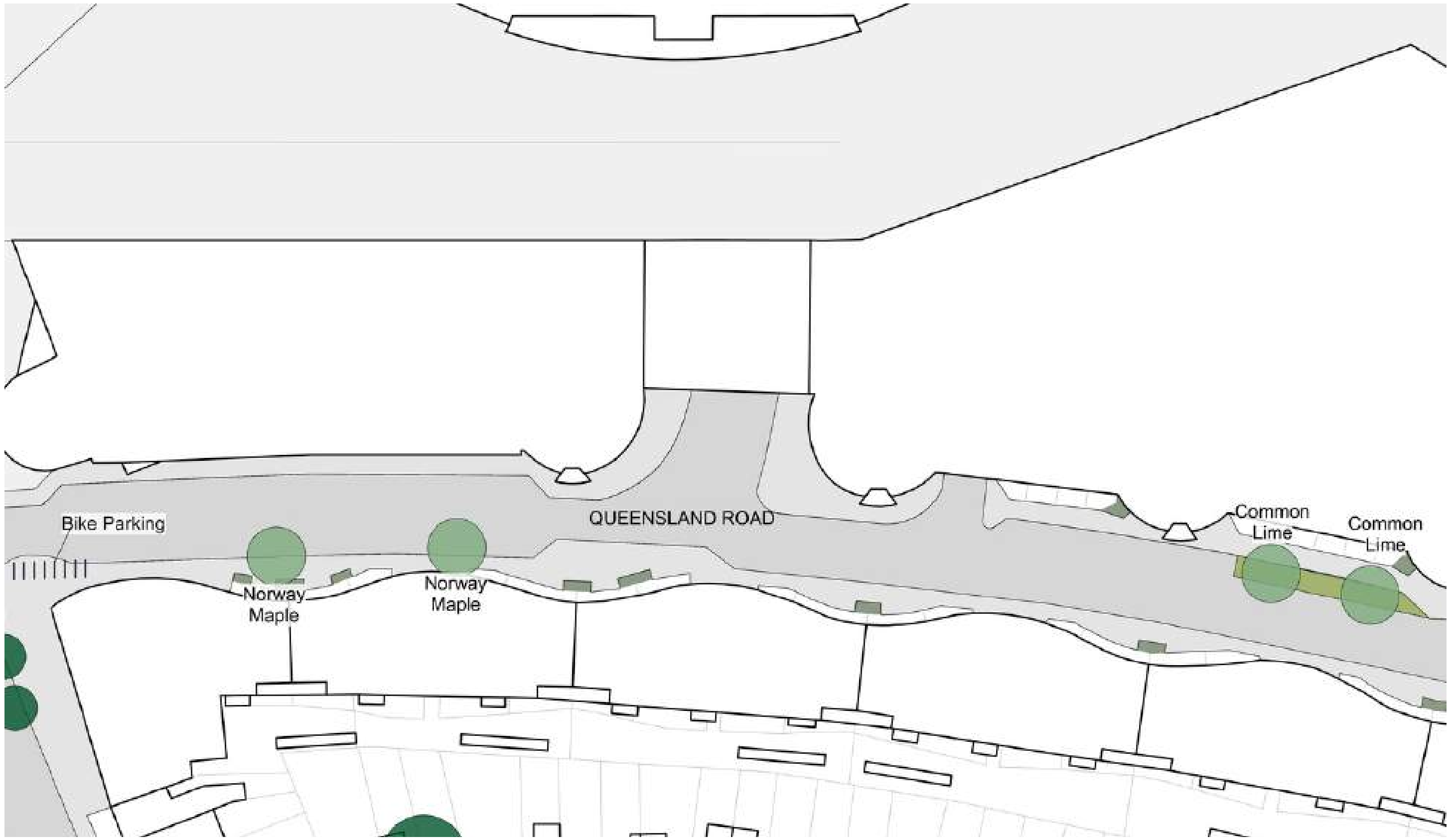
90% Nature
10% People

Design Components

BENEFITS

Social and health: Bike parking, enhanced streetscape.

Climate and environment: Reduces flooding.



Existing Tree New Tree

Figure 72. Queensland Road proposed design

UGF: 0.23

CARBON STORAGE	CARBON SEQUESTERED	RUNOFF AVOIDED	LINEAR TREE CANOPY	PM 2.5 REDUCED	NO2 REDUCED
1.903 tonnes	0.049 tonnes/year	2.48 m3/year	64 m	0.036 kg/yr	0.598 kg/yr

7 Climate and Health Benefits and Limitations

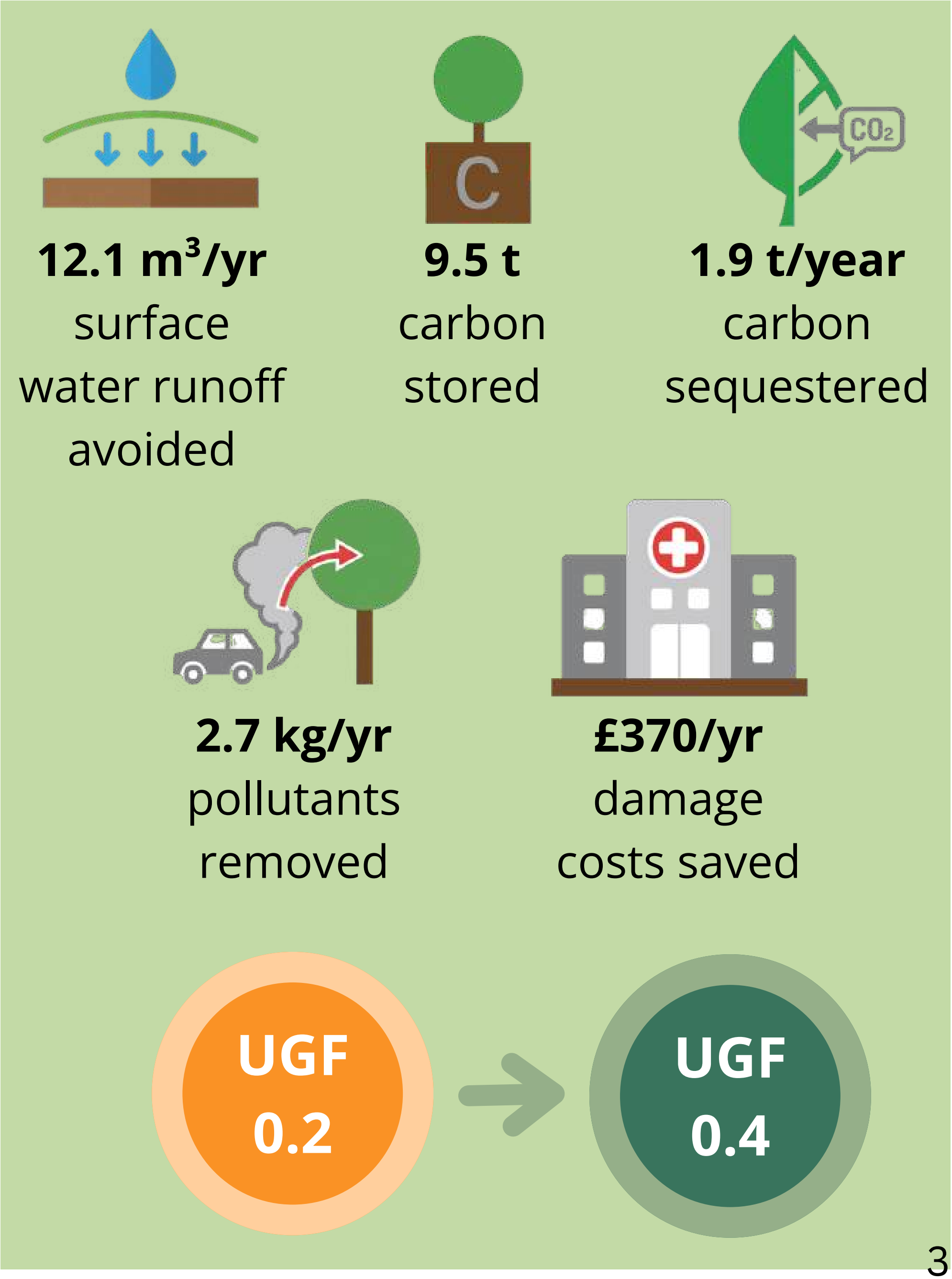
7.1 Climate and Health Benefits

Quantified benefits

The Proposed pocket parks were designed with local-level climate and health risks in mind, and their implementation would bring multiple benefits to Arsenal. The project would introduce around 2100 m^2 of new green space across the 10 sites, consisting of: 1000m^2 of tree canopy cover from 28 trees; 746m^2 of grass verges; 200m^2 of flower and community garden planting; 129m^2 of rain gardens (Appendix Table 1). It's estimated that the addition of these 28 trees would equate to 12.1m^3 of surface water runoff avoidance per year, 9.5 tonnes of carbon stored, and 1.9 tonnes of carbon sequestered each year. Approximately 2.7 kilograms of pollutants (2.5 kg of NO2 and 0.2 kg PM2.5) would be removed from the atmosphere each year, which provides co-benefits to both climate and residents' health. Calculations show that the NHS could save around £370 each year in 'avoided health costs' through reduced GP visits and hospital admissions. See Appendix Table 2 for more in-depth estimates. The average UGF across the 10 sites increased from 0.2 to 0.4 after the pocket park greening interventions, which increased canopy coverage and permeability of surfaces. This is a notable difference, and meets the Mayor's recommended target score of 0.4 for developments in residential areas (Greater London Authority, 2021).

Limitations

It is important to acknowledge that these benefits are calculated using values which represent fully grown trees. Many of the pocket parks proposed in the project involve planting saplings, and it will take many years to calculate climate and health benefits. Health benefit calculations use pollutant removal estimates which represent averages across all London tree species, failing to acknowledge the vast differences in pollutant removal among different species. Additionally, these values were taken by visually estimating from the bar graphs in the Islington i-Tree Eco Inventory Report, introducing a level of error to the calculations. One of the main limitations of the calculated health benefits is that it only considers the influence trees have on improving local air quality, while ignoring potential negative effects of allergenic tree species, which are known to increase local asthma hospitalisation rates (Lai and Kontokosta, 2019). Moreover, trees naturally emit Biogenic volatile organic compounds (BVOCs), which can react with nitrogen oxides (NOx) from vehicles to produce ground-level ozone, a harmful respiratory pollutant (Maison et al., 2024). These negative effects are especially prominent in regions of high pollution, requiring even more careful selection of trees according to local-level pollutant emissions.



7.2 'Hidden' benefits

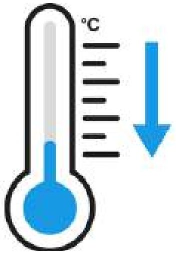
Biodiversity

The impact of pocket parks on biodiversity specifically is not well researched. Quantifying biodiversity benefits requires costly, long-term assessments, which are usually prioritised for larger-scale interventions like parks. However, pocket parks create new opportunities to increase habitat connectivity, allowing species to move more efficiently through fragmented green spaces within urban environments (Fairbrass et al., 2018). This can be achieved at smaller scales, including grass verges, which, if planted with suitable vegetation, can provide valuable habitats for pollinators (Fairbrass et al., 2018). The Urban Bee Paradise proposed for Lowman Road is the perfect example of this. Future pocket parks could further leverage their biodiversity benefits by adding artificial biodiversity habitat structures such as bird, bat and insect boxes (Fairbrass et al., 2018). Increased biodiversity is associated with greater ecological resilience to disturbances like pollution, pests and diseases (London Assembly, n. d.).

Mental and general health

Numerous studies have shown that exposure to urban green spaces and infrastructure, consistently improves mental health and psychological wellbeing (Fairbrass and Washbourne, 2018). Additionally, measures of cognitive function such as attentional capacity and memory, are higher in natural environments (Kondo et al., 2018). Children in greener neighbourhoods display fewer symptoms of anxiety and depression, and those with ADHD experience milder symptoms when playing in green settings (Natural England, 2025). Finally, residents' accesibility to spaces has been linked to increases in physical activity and providing protection against chronical conditions (such as coronary heart disease, asthma and diabetes) (Fairbrass and Washbourne, 2018; Kondo et al., 2018).

Temperature



Pocket parks often create their own microclimates, which is especially important in the face of intense urban heat island effects. Changes such as air temperature, humidity and solar radiation are often extremely modest, and some studies even found higher temperatures in pocket parks relative to their surroundings during midday (Russo et al., 2022; Russo et al., 2024). However, the same studies found that urban pocket parks significantly enhance the improvement of subjective comfort, shifting perspectives from neutral/bad to good/very good (Russo et al., 2022; Russo et al., 2024). This discrepancy highlights the need for better design for microclimate mitigation of pocket parks to fully optimise the benefits of heat-related comfort.

Social

Pocket parks act as community gathering spaces, facilitating social interactions between residents. Drawing together members of the community, pocket parks can help reduce crime and create a sense of safety (Fairbrass and Washbourne, 2018). Play spaces (such as the 'adventure playground' on Citizen Road II) community gardens (on Citizen Road I) and resident-led art spaces (on Annette road) are just a few examples of how pocket parks provide social benefits to residents of all ages.

8 Conclusion

8.1 Conclusion

The benefits of urban greening have been highly prioritised within greater London in recent years, as seen in Islington's 2023 Local Plan and the borough's Liveable Neighbourhoods Initiative. However, the benefits of urban greening interventions have rarely been quantified (Turcu, 2025). This report proposes urban greening interventions designed around the demographics and climate vulnerabilities of Arsenal to maximise their hidden and co-benefits.

The socio-economic inequality of Arsenal presents a unique opportunity to address unequal green space access as well as climate and health risks for deprived residents. The clusters of pocket park interventions proposed throughout this report maximise impact and increase green equity in Arsenal by strategically targeting areas with overlaps of vulnerable residents and high climate risks. The environmental benefits of each design were quantified to demonstrate the multifunctionality and impact of greenspace, crucial to building political and community support for implementation. Additionally, proposed designs catalyse underutilised community networks and groups in Arsenal to create a framework of greenspace stewardship for long-term management.

Calculations show that the small-scale nature of pocket parks, despite their intent, unlock only minimal transformative environmental benefits without scaling across the whole of London. These small-scale interventions address the symptoms, not the root causes of a warming planet. However, the hidden social benefits of these pocket parks should not be ignored. Urban greening strategies that prioritise implementation in areas with higher concentrations of deprived and vulnerable residents create co-benefits that are both environmental and social. The designs found within this report highlight the importance of co-benefits and can be leveraged by municipalities looking to implement pocket park implementation strategies as part of their urban greening initiatives.

Figure 72. Pocket Park in Arsenal

Figure 73. Emirates Stadium

Figure 74. London Overground Rails

Figure 75. Harvist Estate

Figure 76. Victorian Terrace in Arsenal

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All figures are the authors' own unless otherwise indicated.
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10 Appendix

Table 1. Values used for calculating the Urban Greening Factor (UGF) for 10 pocket park sites in Arsenal

				Area (m^2)					
Site	Total site area (m^2)	Intervention Stage	Urban Greening Factor	Standard trees in connected tree pits (min soil volume >= 2/3 of projected canopy area)	Flower-rich perennial planting	Rain gardens and other vegetated sustainable drainage elements	Standard trees planted in pits (soil volume < 2/3 projected canopy area)	Groundcover planting	Amenity grassland (species-poor, regularly mown lawn)
Lowman Road	519	Old	0.1	-	-	-	73	-	-
		Proposed	0.43	163	16	-	-	-	237
		Difference	0.33	163	16	-	-73	-	237
Annette Road	447	Old	0.46	-	111	-	216	-	-
		Proposed	0.46	-	111	-	216	-	-
		Difference	0	-	-	-	-	-	-
Dunford Road	414	Old	0.1	-	-	-	67	-	-
		Proposed	0.54	113	20	-	67	-	200
		Difference	0.44	113	20	-	-	-	200
Citizen Road I	200	Old	0.36	48	-	-	-	-	86
		Proposed	0.38	48	-	-	-	26	60
		Difference	0.02	-	-	-	-	26	-26
Citizen Road II	741	Old	0.4	123	-	-	-	-	505
		Proposed	0.43	123	63	-	-	-	442
		Difference	0.03	-	63	-	-	-	-63
Plimsoll Road	326	Old	0.05	-	-	-	29	-	-
		Proposed	0.44	-	84	-	141	-	-
		Difference	0.39	-	84	-	112	-	-
Monsell Road	929	Old	0.2	129	-	-	133	-	-
		Proposed	0.28	-	-	129	276	-	-
		Difference	0.08	-129	-	129	143	-	-
Chatterton Road	848	Old	0.1	-	-	-	147	-	-
		Proposed	0.36	-	-	-	424	-	130
		Difference	0.26	-	-	-	277	-	130
Elwood Street	530	Old	0	-	-	-	-	-	-
		Proposed	0.53	242	-	-	-	-	211
		Difference	0.53	242	-	-	-	-	211
Queensland Road	2148	Old	-	-	-	-	-	-	-
		Proposed	0.08	-	-	-	228	-	57
		Difference	0.08	-	-	-	228	-	57
Net area of additional cover	**7102**			**389**	**183**	**129**	**687**	**26**	**746**

Table 2. Calculations for Health Benefits of 10 pocket parks in Arsenal, Islington

				Climate Benefits						Health Benefits			
Cluster	Site	Typology	Number of trees added	Carbon Storage (t)	Carbon Sequestration (t/yr)	Runoff avoided (m3/yr)	pollution avg (kg/yr)	PM2.5 removal (kg/yr)	NO2 removal (kg/yr)	Damage Costs Saved (£/yr)	Life years lost (ly/yr)	Annual hospital admissions prevented	Lifetime value (50 years)
I	Lowman Road	Bee Paradise	3	1.32	0.04	1.245	0.63	0.02	0.23	94.33	0.00160	0.0000130	3236
	Annette Road	Urban Art Space	-	-	-	-	-	-	-	-	-	-	-
	Dunford Road	Social Gardening	2	0.09	0.40	1.24	0.63	0.02	0.30	32.79	0.00006	0.0000017	1125
II	Citizen Road I	Social Gardening	2	0.14	0.00	0.09	0.01	0.01	0.01	6.92	0.00001	0.0000003	237
	Citizen Road II	Spontaneous Play	3	2.75	0.71	2.17	1.02	0.03	0.35	43.92	0.00008	0.0000024	1506
III	Plimsoll Road	Active Transport Landscape	2	0.70	0.02	0.89	0.23	0.01	0.23	23.73	0.00004	0.0000071	814
	Monsell Road	Active Transport Landscape	2	0.74	0.01	0.99	0.22	0.01	0.24	26.2	0.00005	0.0000077	899
	Chatterton Road	Active Transport Landscape	7	1.14	0.03	1.38	2.01	0.09	0.35	54.22	0.00037	0.0000392	1813
	Elwood Street	Play on the Way	3	0.73	0.40	1.59	0.70	0.01	0.23	21.96	0.00004	0.0000012	753
Outlier	Queensland Road	Active Transport Landscape	4	1.90	0.26	2.48	1.20	0.04	0.60	65.46	0.00012	0.0000035	2245
Total			**28**	**9.51**	**1.87**	**12.08**	**6.65**	**0.24**	**2.54**	**369.53**	**0.00237**	**0.0000762**	**12628**